2021-2022 Spring Term / Strength of Materials II/ Homework

Start Date: 20.05.2022 Friday 21:00

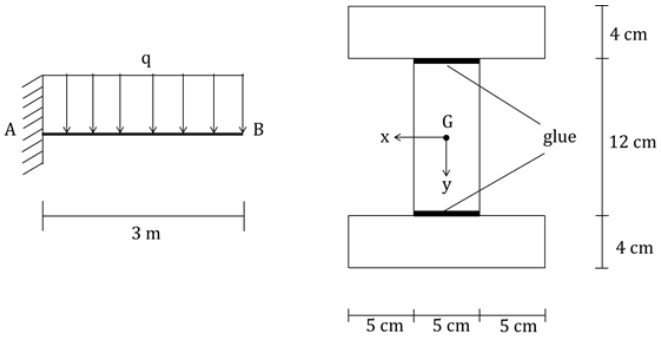
End Date: 27.05.2022 Friday 21:00

Name-Surname:

Group:

Number:

1- A wooden beam having a span length of 3m is built with three timber parts connecting by glue and supported as shown in the figure. Knowing that the allowable stresses for the material of the beam are $\tau_{all} = 0.8\,MPa$, $\sigma_{all} = 14\,MPa$ and safety shear stress for the glue is $\tau_{all,glue} = 0.4\,MPa$,



a- Find the load **q** that can be safely carried by the beam,

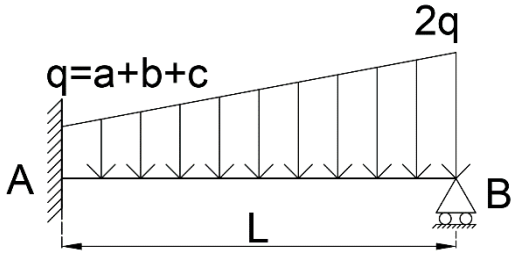
b- Using the **q** value you obtained, draw the normal and shear stress diagrams in the most critical section of the beam.

2- For the beam given in the figure,

a- Find the equation of elastic curve by **integration method**.

b- Draw (T_y), (M_x) diagrams of the beam.

c- Find the support reactions.



Explanation:

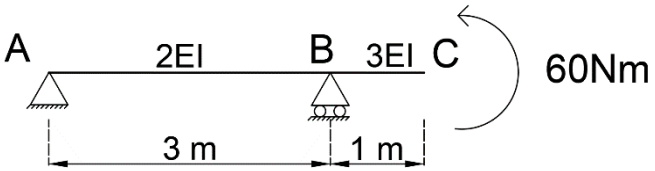
Number a: is the last digit of your student number.
Number b: is the second last digit of your student number.
Number c: is the third last digit of your student number.

- Example for **"q=a+b+c (N)":** Student number: xxxxx394, *a*=4, *b*=9, c=3 then **q=4+9+3=16 N**
- Example for **"L=1ab (cm)":** Student number: xxxxxx94, *a*=4, *b*=9 then **L=149 cm**

3- For the beam given in the figure,

a- Draw the elastic curve by using the **Mohr Method**.

b- Calculate the values of Θ_A, Θ_B, V_C, Θ_C by using the **Mohr Method**.



4- The cross-section of the ABCD element in the figure is a circle with a diameter of 6 cm. The force of P=500 N is in the x-z plane. The CD segment is parallel to the y-axis. In line with the information given,

a- Find the most critical points in section A.

b- Calculate the normal and shear stresses at the most critical points in section A.

c- Calculate the principal stresses at the specified points.

d- Check the section with the Tresca Hypothesis.
$\sigma_{em} = 95\,MPa$

